F12 7th NRN

Lecture 24: Section 3.5
Exponential and Logarithmic Models

Exponential growth model

Exponential decay model

Logistic growth model

Doubling time/half-life time

How is b derived if doubling time is know?

$y = ae^{bt}$

Let y double then $y = 2a$

let $t = T$ since doubling time.

$\frac{1}{a}(2a = ae^{bT})$

solve for b

$2 = e^{bT}$

$\ln 2 = \ln e^{bT}$

$\ln 2 = bT \underbrace{\ln e}_{=1}$

$(\ln 2 = bT)\frac{1}{T}$

$b = \frac{\ln 2}{T}$

Lecture 24: Section 3.5 Exponential and Logarithmic Models

Four common types of mathematical models involving exponential functions and logarithmic functions.

usually your first step is to find b.

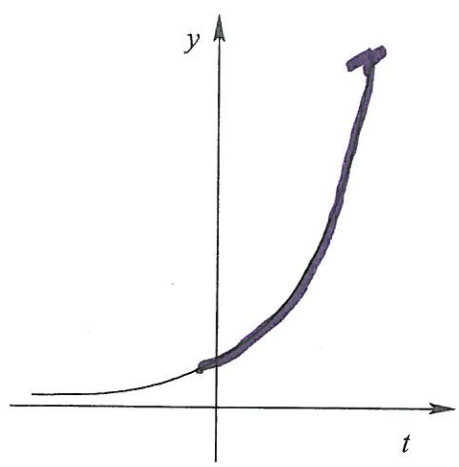
1. Exponential growth model: $\boxed{y = ae^{bt}}$

where y = amount at time t

a = amount at time 0, initial amt.

b = relative growth rate, $b > 0$

t = time



NOTE: If the doubling time T is given, then

$$b = \frac{\ln 2}{T}$$

$y = ae^{bt}$

we know $a = 900$
find b.

The population of a small town grows exponentially. In 1900, the population was 900. In 1920, the population had grown to 1,600.

1) What is the growth rate of the population of this town?

year	t	y = population
1900	t=0	900
1920	t=20	1600

solve for b

$y = ae^{bt}$

$1600 = 900e^{b(20)}$

$\frac{16}{9} = e^{20b}$

$\ln\frac{16}{9} = \ln e^{20b}$

$\ln\frac{16}{9} = 20b \ln e$ $= 1$

$e^{\ln x} = x$

2) What was the population of this town in 1960?

Let $t = 60$

$y = 900e^{\frac{\ln\frac{4}{3}}{10}(60)}$ 6

$y = 900e^{6\ln\frac{4}{3}}$

$y = 900e^{\ln\left(\frac{4}{3}\right)^6}$

$y = 900\left(\frac{4}{3}\right)^6 \approx 5057$ people

$\frac{\ln\frac{16}{9}}{20} = b$

$\frac{\ln\left(\frac{4}{3}\right)^2}{20} = \frac{2\ln\frac{4}{3}}{20} =$

$\frac{\ln\frac{4}{3}}{10} = b \approx 0.0288$

Since the doubling time is known $b = \frac{\ln 2}{T}$

$b = \frac{\ln 2}{40}$

ex. A culture starts with 10,000 bacteria, and the number doubles every 40 minutes.

First find a and b: $y = ae^{bt}$

1) Find a function that models the number of bacteria at time t. $a = 10{,}000$ $b = ?$

$$y = 10{,}000\, e^{\frac{\ln 2}{40} \cdot t}$$

$e^{\ln x} = x$

2) Find the number of bacteria after ~~one~~ two hours. Let $t = 120$ min

$y = 10{,}000\, e^{\frac{\ln 2}{\cancel{40}} \cdot (\cancel{120}\,3)} = 10{,}000\, e^{3\ln 2} = 10{,}000\, e^{\ln 2^3}$

$y = 10{,}000(2^3) = 80{,}000$ bacteria

3) After how many minutes will there be 50,000 bacteria? let $y = 50{,}000$, find t.

$50{,}000 = 10{,}000\, e^{\left(\frac{\ln 2}{40}\right)t}$

$5 = e^{\left(\frac{\ln 2}{40}\right)t}$

$\ln 5 = \ln e^{\left(\frac{\ln 2}{40}\right)t}$

$\ln 5 = \left(\frac{\ln 2}{40}\right)t \cdot \underbrace{\ln e}_{=1}$

$\left[\ln 5 = \frac{\ln 2}{40} \cdot t\right] \frac{40}{\ln 2}$

$t = \ln 5\left(\frac{40}{\ln 2}\right) =$

$t = 40\left(\frac{\ln 5}{\ln 2}\right) \approx 93$ min.

How to derive $b = \frac{\ln 2}{T}$ if half-life is given:

let $\frac{a}{2} = y$ = half of the amt.

let $t = T$ since half-life

2. Exponential decay model: $y = ae^{-bt}$

where y = amount at time t

a = amount at time 0

b = relative decay rate, $b > 0$

t = time

$$\frac{1}{a}\left(\frac{a}{2} = ae^{-bT}\right)$$

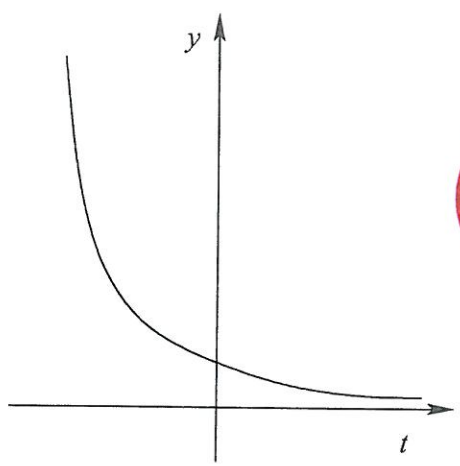
$$\frac{1}{2} = e^{-bT}$$

$$\ln\frac{1}{2} = \ln e^{-bT}$$

$$\ln 2^{-1} = -bT \ln e$$

$$\left(-1\ln 2 = -bT\right)\left(-\frac{1}{T}\right)$$

$$\frac{\ln 2}{T} = b$$



NOTE: If the half life T is given, then

$$b = \frac{\ln 2}{T}$$

$b = \frac{\ln 2}{T}$ if half-life is given

ex. The radioactive isotope gallium 67 (^{67}Ga), used in the diagnosis of malignant tumors, has a biological half-life of 46.5 hours. $y = ae^{-bt}$

1) If we start with 100 milligrams of the isotope, then find a function that models the amount of the isotope remaining at time t. Find a and b. $a = 100$

$b = \frac{\ln 2}{46.5}$

$y = 100e^{-\frac{\ln 2}{46.5} \cdot t}$

2) How many milligrams will be left after 24 hours?

let $t = 24$

$\boxed{y = 100e^{\left(-\frac{\ln 2}{46.5}\right)(24)}} \approx 69.9$ mg

$y = \log_a x$

$(1,0)$

<u>ex.</u> Strontium 90 is a radioactive material that decays according to the model $y = ae^{-0.0244t}$. Assume that a scientist has a sample of 500 grams of strontium 90.

$y = ae^{-bt}$ $\quad$ $y = 500e^{-0.0244t}$

1) What is the decay rate of strontium 90?

$b = 0.0244$

2) How much strontium 90 is left after 10 years? let $t = 10$

note: $a = 500$

$y = 500e^{-0.0244(10)}$

$y = 500e^{-0.244}$ grams ≈ 391.74 grams

3) When will 400 grams of strontium 90 be left? Let $y = 400$ solve for t.

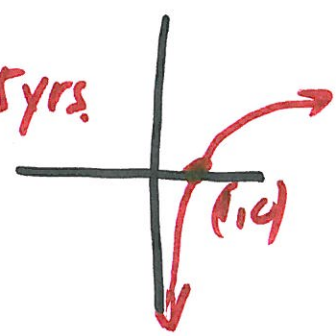
$400 = 500e^{-0.0244t}$

$\frac{4}{5} = e^{-0.0244t}$

$\ln\frac{4}{5} = \ln e^{-0.0244t}$

$\ln\frac{4}{5} = -0.0244t$

$t = \frac{\ln\frac{4}{5}}{-0.0244} \approx 9.15$ yrs.



4) What is the half-life of strontium 90?

Take $\frac{1}{2}(500) = 250$ grams Let $y = 250$ find t.

$250 = 500e^{-0.0244t}$

$\frac{1}{2} = e^{-0.0244t}$

$\ln\frac{1}{2} = \ln e^{-0.0244t}$

$\ln 2^{-1} = -0.0244t$

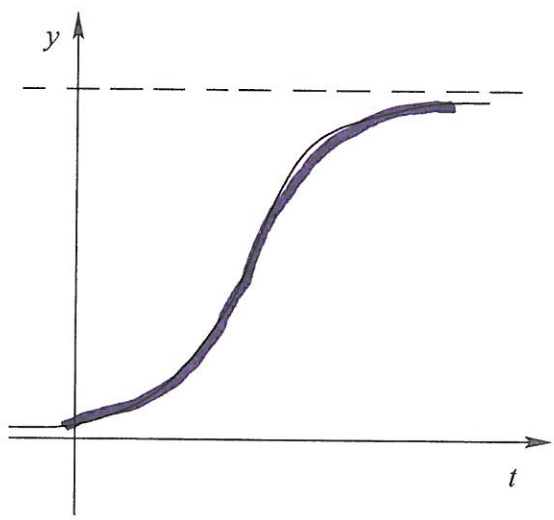
$t = \frac{\ln 2^{-1}}{-0.0244} = \frac{+\ln 2}{+.0244} \approx 28$ years

as $t \to \infty$

$e^{-rt} \approx e^{-\infty} = \frac{1}{e^{\infty}} \to 0$ as $t \to \infty$

3. Logistic growth model: $y = \dfrac{a}{1 + be^{-rt}}$ ≈ 0 as $t \to \infty$

where y = population at time t, and $a, b, r > 0$.



$y = \dfrac{a}{1 + b(0)} = \dfrac{a}{1}$

$= a$

NOTE:

1. Short-term population growth uses the exponential growth models, and long-term population growth uses logistic growth models.

2. The number a is called the carrying capacity because y approaches a as t approaches infinity; that is, $\lim_{t \to \infty} y = a$.

ex. Fruit flies are placed in a half-pint milk bottle with a banana and yeast plants (for food and to provide a stimulus to lay eggs). Suppose that the fruit fly population y after t days is given by

$$y(t) = \frac{240}{1 + 79e^{-0.5t}}$$

1) How many fruit flies were initially placed in the bottle? $t=0$

$$y(0) = \frac{240}{1+79e^{0}} = \frac{240}{1+79} = \frac{240}{80} = 3 \text{ flies}$$

2) When will the population of fruit flies be 80? Let $y=80$ solve for t.

$$80 = \frac{240}{1+79e^{-0.5t}}$$

$$80(1+79e^{-0.5t}) = 240$$

$$1+79e^{-0.5t} = 3$$

$$79e^{-0.5t} = 2$$

$$e^{-.5t} = \frac{2}{79}$$

$$\ln e^{-.5t} = \ln \frac{2}{79}$$

$$-.5t = \ln \frac{2}{79}$$

$$t = \frac{\ln \frac{2}{79}}{-.5} \text{ days}$$

≈ 7 days

3) What is the population of fruit flies as $t \to \infty$?

240

ex. The logistic growth model

$$P(t) = \frac{0.90}{1 + 3.5e^{-0.339t}}$$

relates the proportion of new personal computers sold at Best Buy that have Intel's latest coprocessor t months after it has been introduced.

1) What proportion of a new personal computers sold at Best Buy will have Intel's latest coprocessor when it is first introduced? Let $t=0$

$$P(0) = \frac{0.90}{1+3.5e^0} = \frac{.90}{4.5} = \frac{.9}{4.5}\left(\frac{10}{10}\right) = \frac{9}{45} = \boxed{\frac{1}{5}}$$

2) Determine the maximum proportion of new personal computers sold at Best Buy that will have Intel's latest coprocessor. As $t \to \infty$

$$\frac{1}{e^{.339t}} \to \frac{1}{\text{big \#}} \approx 0$$

$$\frac{.90}{1+3.5\left(\frac{1}{e^{.339t}}\right)} \approx \frac{.9}{1+3.5(0)} = \frac{.9}{1} = \boxed{\frac{9}{10}}$$

or see note #2 on page 7.

3) When will 75% of new personal computers sold at Best Buy have Intel's latest coprocessor?

$$.75 = \frac{.9}{1+3.5e^{-.339t}}$$

$$.75(1+3.5e^{-.339t}) = .9$$

$$1+3.5e^{-.339t} = \frac{.9}{.75}$$

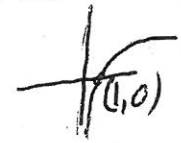
$$3.5e^{-.339t} = \frac{.90}{.75} - 1$$

$$e^{-.339t} = .2$$

$$\ln e^{-.339t} = \ln .2$$

$$-.339t = \ln .2$$

$$\boxed{t = \frac{\ln .2}{-.339} \approx 4.75 \text{ months}}$$



ln .2 is neg

$$\frac{.90}{.75} = \frac{.18}{.15} = \frac{6}{5} = 1.2$$